

Short Communication

Children's motor imagery modality dominance modulates the role of attentional focus in motor skill learning

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ABSTRACT

We investigated whether children's motor imagery dominance modulated the relationship between attentional focus and motor learning of a tossing task. One hundred and thirty-eight boys (age: $M = 10.13$, $SD = 0.65$) completed the Movement Imagery Questionnaire – Children (MIQ-C) to determine imagery modality dominance (kinesthetic, internal-visual, external-visual) and were randomly assigned to either an internal ($n = 71$) or external ($n = 67$) attentional focus group. Participants completed 60 trials of a tossing task with their non-dominant hand on day 1. Participants in the internal focus group were asked “to focus on the throwing arm”, whereas participants in the external focus group were instructed “to focus on the ball.” A retention test was conducted 24 h later to assess motor learning. Overall, the results from a nested, multiple linear regression analysis indicated the degree to which internal or external focus influences children's throwing accuracy is dependent upon their motor imagery modality dominance. Specifically, higher levels of external-visual imagery dominance resulted in greater motor learning for children adopting an external focus. In contrast, higher values of kinesthetic imagery dominance resulted in reduced motor learning for children who adopted an external focus. Despite the need for future research, we recommend motor imagery modality dominance assessments be considered when investigating the influence of attentional focus on motor learning, particularly when the target population is children.

1. Introduction

Motor imagery is defined as a cognitive process that involves imagining a movement without physical involvement (Mulder, 2007). Importantly, the ability to clearly image one's movements has been found to influence the performance and learning of motor tasks in

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both children and adults (Cumming & Williams, 2012; Goss, Hall, Buckolz, & Fishburne, 1986; Mizuguchi, Nakata, Uchida, & Kanosue, 2012; Robin et al., 2007; Williams, Thomas, Maruff, & Wilson, 2008). To this extent, research has shown that children with motor impairments can have difficulty in performing motor imagery tasks, suggesting a direct link between motor imagery and motor performance (Williams et al., 2008). As such, evidence indicates that motor imagery dominance is an influential factor for skill acquisition and learning.

Research on motor imagery and skill acquisition has distinguished between two common types of motor imagery modalities: visual and kinesthetic (Féry, 2003; Guillot, Collet, & Dittmar, 2004). This is an important distinction as individual characteristics (Guillot et al., 2004) and environmental/task constraints (Guillot & Collet, 2004) influences whether an individual exerts dominance in one of these two modalities. Individuals with visual imagery dominance are better at visualizing movements in their mind, meaning they are more apt at 'seeing' certain surface features such as space and size. Visual imagery can be further sub-divided into internal- and external visual imagery. Internal-visual imagery involves visualization of the movement from the performer's visual perspective (i.e., first-person view), whereas, external-visual imagery allows individuals to visualize themselves from a third-person perspective (Yu et al., 2016). In contrast, individuals with kinesthetic imagery dominance are more attuned to sensing movement and body limb position, such as feeling the sense of force (e.g., during a jump) or feeling the movement of the arms (e.g., during a dart throw) (Guillot et al., 2009). Importantly, differences in imagery modality dominance can influence the rate of skill acquisition. For example, individuals with high visual/high kinesthetic imagery dominance acquire movement patterns faster than high visual/low kinesthetic imagers (Goss et al., 1986) and external-visual imagery is more effective than internal-visual imagery in the learning of a new kata (Hardy & Callow, 1999).

Aside from motor imagery modality dominance, attentional focus has also been shown to play a critical role in the performance and learning of motor skills. In motor behavior literature, how one allocates their limited cognitive resources has been distinguished between internal and external foci of attention (Wulf, 2013). An external focus of attention is typically defined as concentration on external task-relevant information or one's intended movement effects (object-related information/desired movement outcomes; Wulf, Höß, & Prinz, 1998). In contrast, an internal focus of attention is defined as concentration on internal body-relevant movements (body-related information; Wulf et al., 1998). Relative to an internal focus of attention, an external focus of attention robustly supports enhanced motor performance and motor learning across a variety of tasks in adults, children, and special populations (for reviews see Wulf, 2013 and Wulf & Lewthwaite, 2016). However, an emergent line of research has revealed that the relative effects of attentional focus instructions on motor behavior may depend on participants' dominance in a given motor imagery modality (visual or kinesthetic; Bahmani, Diekfuss, & Kharestani, 2019; Sakurada, Hirai, & Watanabe, 2016, 2019; Sakurada, Nakajima, Morita, Hirai, & Watanabe, 2017).

The seminal paper exploring the interaction between motor imagery modality dominance and attentional focus asked healthy individuals with either visual or kinesthetic imagery dominance to perform a visuomotor rotation task using both an internal and external focus of attention (Sakurada et al., 2016). Results demonstrated that individuals with visual imagery dominance performed greater when adopting an external focus; whereas individuals with kinesthetic imagery dominance performed superior when adopting an internal focus. These findings were subsequently replicated in patients recovering from an acute stroke (Sakurada et al., 2017). Follow-up studies using functional near-infrared spectroscopy (fNIRS) have further revealed that, when participants adopt an attentional focus that corresponds to their motor imagery dominance (i.e., external focus for visual imagery dominant; internal focus for kinesthetic imagery dominant), the relative behavioral improvements were associated with more efficient brain activity (Sakurada et al., 2019). Consequently, the authors suggested that the optimal attentional focus strategy is dependent on an individual's dominant motor imagery modality (Sakurada et al., 2016, 2019, 2017).

The studies by Sakurada and colleagues, however, manipulated participants' attention by directing it towards aspects of fine motor control tasks (e.g., hand movement/cursor movement on screen), which can elicit differential cognitive demands when compared to attention directed towards aspects of gross motor control (Raisbeck & Diekfuss, 2015). One study did examine how visual and kinesthetic imagery dominance interacted with attentional focus instruction using gross motor tasks (balance, experiment 1; dart throwing, experiment 2), but did not fully replicate previous findings (Bahmani, Diekfuss, & Kharestani, 2019). Specifically, results from both experiments indicated that the visual imagery dominance group performed (experiment 1) and learned (experiment 2) better with an external relative to an internal focus of attention. However, there were no significant differences in performance or learning, when adopting an internal or external focus of attention, for those with kinesthetic imagery dominance (Bahmani, Diekfuss, & Kharestani, 2019). In addition to the differences in tasks, experiment 2 elicited a longer retention period (24 h) than the studies by Sakurada and colleagues (same day visits), potentially indicating that attentional focus effects were not retained for those with high kinesthetic imagery dominance. As such, the extant literature supporting the optimal attentional focus strategy based on participants' imagery dominance is still unclear, particularly for those who are kinesthetic dominant. Considering kinesthetic-related information is particularly vital for motor learning during the early stages of skill acquisition (Peh, Chow, & Davids, 2011; Pinzon, Vega, Sanchez, & Zheng, 2017)—with previous studies all conducted using adult participants—examining the potential interaction between attentional focus and motor imagery modality dominance in children could help clarify previous findings.

Exploring such an interaction could also help to clarify previous literature examining the effects of attentional focus for motor learning in children. In particular, while several studies have indicated the beneficial learning effects for children adopting an external focus (Abdollahipour, Palomo Nieto, Psotta, & Wulf, 2017; Abdollahipour & Psotta, 2017; Brocken, Kal, & van der Kamp, 2016; Perreault & French, 2015), there are some studies that did not find such advantages (e.g., Chow, Koh, Davids, Button, & Rein, 2014; Emanuel, Jarus, & Bart, 2008; Perreault & French, 2016). Although methodological differences may have played a role in these divergent findings, another plausible reason could be related to children's motor imagery modality dominance (i.e., visual or kinesthetic). Imagery modality dominance has been shown to develop throughout childhood (Spruijt, Jongsma, van Der Kamp, &

Steenbergen, 2017), and has yet to be considered in previous literature. Therefore, the purpose of our study was to examine if children's motor imagery modality dominance influenced the relationships between attentional focus and motor learning of an overarm ball-tossing task. Consistent with the majority of past research, we hypothesized that children with higher values of kinesthetic imagery dominance would elicit superior learning when provided internal focus instructions, whereas children with higher values of visual imagery dominance (including internal- and external-visual imagery) would demonstrate superior learning when provided external focus instructions.

2. Method

2.1. Participants

One hundred and thirty-eight boys¹ ($Mage = 10.13$, $SD = 0.65$ years) completed this study. Participants had no prior experience with the task and were naïve as to the purpose of the study. The study was approved by the university's institutional review board. Prior to data collection, parents of the participants signed an informed consent form, and assent was verbally obtained from participants. An a priori power analysis indicated that 119 participants would be sufficient to identify significant multiple regression effects using a model with eight predictor variables (model 4 described below) and power ($1 - \beta$) of 0.95, effect size f^2 of 0.15 and an α of 0.05 (Faul, Erdfelder, Lang, & Buchner, 2007).

2.2. Apparatus and task

Participants performed an overarm toss with a tennis ball using their non-dominant hand to a target at a distance of 20 ft (6.1 m). The target was hung on a wall and consisted of 10 concentric circles with a bull's eye in the center. The bull's eye had a 10-cm radius surrounded by concentric circles with radii of 20, 30, 40, 50, 60, 70, 80, 90, 100 cm, respectively. The center of the bull's eye was 4.82 ft (1.47 m) above the ground. We adopted a common scoring system for aiming-related tasks studying motor performance and learning. (e.g., Bahmani, Diekfuss, Rostami, Ghadiri, & Ataee, 2018; Diekfuss, Ward, & Raisbeck, 2017; Raisbeck & Diekfuss, 2015, 2017; Wulf, Chiviacowsky, & Cardozo, 2014; Wulf, Lewthwaite, Cardozo, & Chiviacowsky, 2018). Specifically, if the ball hit the bull's eye, 100 points were awarded, 90 points were given if the ball hit the first surrounding circle of the bulls-eye, 80 points were given for balls hitting third circle, and so forth (see Fig. 1). If a toss missed the target, zero points were scored by the experimenter. Toss accuracy was used as the primary dependent variable of interest and determined by dividing an individual's total score in each block by the number of trials completed during one block in each testing phase (pre-test, acquisition, and post-test/retention, respectively; described below). All tosses were recorded using a video camera. Recordings were used if the experimenter was uncertain about where the ball hit with respect to the target.

2.3. Procedure

Prior to performing the tossing task, participants completed the Persian version of the Movement Imagery Questionnaire – Children (MIQ-C; Yavari Kateb, Gharayagh Zandi, & Moghadam Zadeh, 2018). The original and more frequently cited English version of the MIQ-C is a twelve-item, validated questionnaire that assesses 7–12 year old children's motor imagery dominance on three subscales: kinesthetic imagery, internal-visual imagery, and external-visual imagery (Martini, Carter, Yoxon, Cumming, & Ste-Marie, 2016). The original MIQ-C reported fair to substantial test-retest reliability, with intraclass correlation coefficients (ICCs) of 0.82, 0.72, 0.43 for kinesthetic imagery, internal-visual imagery, and external-visual imagery, respectively (Martini et al., 2016). The version we used was identical to the original MIQ-C (i.e., same 12 items), except the questions were translated to Persian (to accommodate the current study participants' primary reading language). The Persian version of the MIQ-C reported moderate to high test-retest-reliability, with ICCs of 0.81, 0.75, and 0.84 for kinesthetic imagery, internal-visual imagery, and external-visual imagery, respectively (Yavari Kateb et al., 2018). Thus, the Persian version of the MIQ-C was considered acceptable for the present investigation; we refer to this version as the 'MIQ-C' henceforth. The maximum score that a child could obtain for each of the MIQ-C subscales was 28.

Next, participants were randomly assigned into internal ($n = 71$) or external ($n = 67$) focus groups. Participants performed the tossing task with their non-dominant hand across two days. On day one, participants performed three familiarization trials (that were not used in the statistical analyses), followed by five pre-test tosses without attentional focus instructions. Next, participants completed six blocks of ten tosses during an acquisition phase (60 total tosses). During acquisition, participants in the internal focus groups were asked "to focus on the tossing arm", whereas participants in the external focus groups were instructed "to focus on the ball." On day two (24 h later), a post-test including 10 tosses was completed to assess motor learning (i.e., 'retention'). No attentional focus instructions were provided before or during the post-test. To determine adherence to the instructions, participants were asked to answer an open-ended question after blocks two, four, and six during acquisition, and following completion of the post-test. The experimenter asked participants the following question: "what were you focusing on while performing your tosses?"

¹ Boys were chosen to eliminate any possible gender differences in motor imagery (Subirats, Allali, Briansoulet, Salle, & Perrochon, 2018).

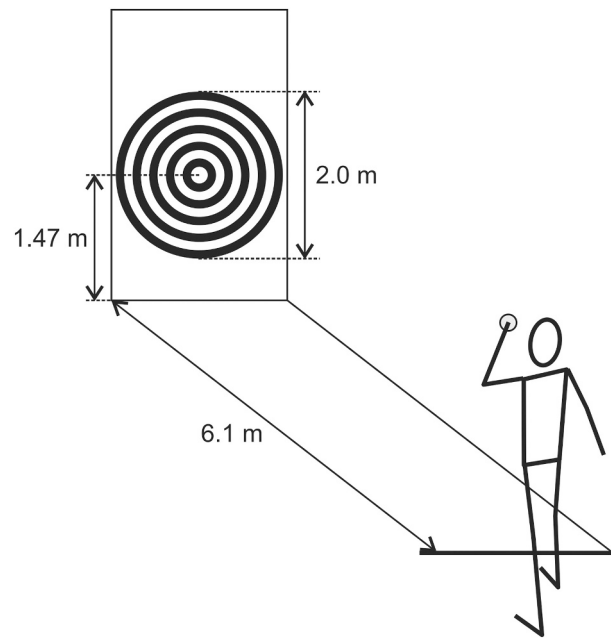


Fig. 1. Schematic of the task.

2.4. Data analyses

As appropriate, descriptive analyses were used for reporting means and standard deviations. Preliminary analyses of the MIQ-C imagery scores identified violations to the assumptions of normality. Therefore, a Friedman test was used to evaluate the differences in mean ranks (M_{Rank}) among imagery dominance modalities. The effect sizes for non-parametric Friedman test was estimated using Kendall's W , with values ranging from 0 (no relationship) to 1 (a perfect relationship). Also, Wilcoxon rank test was used to compare the differences in imagery scores between attentional focus groups. The correlation coefficient r was used to estimate the effect size for Wilcoxon rank test. According to the Cohen's guideline the r values of 0.1, 0.3, and 0.5 correspond to small, medium and large effect size, respectively (Field, 2009).

A nested, multiple linear regression analysis was used to determine the extent to which MIQ-C domain scores and attentional focus instructions predicted motor learning (i.e., toss accuracy scores during the *post-test* with no attentional focus instruction). Model 1 included only the pre-test scores (one total main effect), model 2 included pre-test scores and the internal/external attentional focus group membership (focus group; dummy-coded with 1 = external focus, 0 = internal focus [reference group]; two total main effects), and model 3 included pre-test scores, focus group membership, and each of the MIQ-C subscale measures for kinesthetic imagery, internal-visual imagery, and external-visual imagery as covariates (five total main effects). Model 4 included pre-test scores, focus group membership, each of the MIQ-C subscales, and the interaction between focus group membership and each of the MIQ-C subscale measures. Specifically, model 4 included one main effect of pre-test, one main effect of attentional focus group, three imagery main effects, and three focus group by imagery interaction effects.

Preliminary analyses also indicated all appropriate statistical assumptions were met, as no violations to the assumptions of normality, linearity, or multicollinearity were present in these data (e.g., variance inflation factors were all 1.3 or below). Significance level was set at $p = .05$. Data were analyzed using IBM SPSS Statistics (version 22.0; IBM, Armonk, NY).

To assess adherence to the attentional focus instructions, the verbal answers of each participant were recorded. Subsequently, the participants' responses were coded into one of three categories (i.e., internal foci, external foci, and other foci). Examples of internal foci were "I was focusing on my tossing arm", and "...my arm movement." Examples of external foci were "I was focusing on the ball", "... on the ball trajectory", and "... on the bull's-eye/target", or "... on the ball trajectory and bull's-eye/target". Responses that did not fit into the aforementioned categories such as "Nothing", "I don't know", "I don't remember" were coded as other foci.

3. Results

3.1. Imagery scores

Means and standard deviations from the three MIQ-C modality subscales for the entire cohort, as well as for each attentional focus group are presented in Table 1.

The results of the Friedman test for all children showed no significant differences in mean ranks among internal-visual ($M_{\text{Rank}} = 1.92$), external-visual ($M_{\text{Rank}} = 2.01$), and kinesthetic ($M_{\text{Rank}} = 2.07$) imagery dominance, $\chi^2(2, N = 138) = 1.757, p = .406$, Kendall's

$W = 0.006$. In particular, for the internal focus group, the results of the Friedman test did not find significant differences in mean ranks among internal-visual ($M_{\text{Rank}} = 1.88$), external-visual ($M_{\text{Rank}} = 1.96$), and kinesthetic ($M_{\text{Rank}} = 2.15$) imagery dominance, $\chi^2(2, N = 71) = 2.891, p = .234$, Kendall's $W = 0.020$. For the external focus group, the results of the Friedman test also did not show significant differences in mean ranks among internal-visual ($M_{\text{Rank}} = 1.96$), external-visual ($M_{\text{Rank}} = 2.06$), and kinesthetic ($M_{\text{Rank}} = 1.99$) imagery dominance, $\chi^2(2, N = 67) = 0.403, p = .810$, Kendall's $W = 0.403$.

The results of Mann-Whitney U test did not find significant differences between the internal and external focus groups in their internal-visual ($M_{\text{Rank-External}} = 75.78, M_{\text{Rank-Internal}} = 63.58, z = -1.798, p = .072, r = 0.153$), external-visual ($M_{\text{Rank-External}} = 65.03, M_{\text{Rank-Internal}} = 73.72, z = -1.280, p = .202, r = 0.109$), and kinesthetic ($M_{\text{Rank-External}} = 68.63, M_{\text{Rank-Internal}} = 70.32, z = -0.248, p = .806, r = 0.021$) imagery sub-scores. Also, the internal and external focus groups did not differ in terms of the overall allocation of individuals with varying imagery preferences ($M_{\text{Rank-External}} = 69.28, M_{\text{Rank-Internal}} = 69.71, z = -0.064, p = .950, r = 0.005$).

3.2. Manipulation check

Analyses of participants' responses to the open-ended question revealed that participants generally followed the instructions. Participants in the external focus group reported that they had focused on either the ball, ball trajectory, or ball's eye/target more frequently. On the other hand, a large number of participants in the internal focus group reported that they had focused on their tossing arm and arm movement. Negligibly, a few participants reported other types of foci that were irrelevant to the task. The percentage of adherence to the attentional focus instructions in acquisition and after the post-test are presented in Tables 2 and 3.

3.3. Toss accuracy

Means and standard deviations of toss accuracy for each attentional focus group during pre-test and post-test are presented in Table 4.

Nested multiple regression analyses were performed to test whether imagery modality predicted toss accuracy of the internal and external groups at post-test (i.e., motor learning/retention; see Table 5). To account for any initial performance differences between the groups, pre-test scores were included as a covariate. There was a statistically significant association between pre-test and post-test accuracy (model 1: Beta = 0.789, 95% CI [0.640, 0.937], $p < .001$). In general, attentional focus instructions was a significant predictor of motor learning in the model unadjusted for imagery modality (model 2: Beta = 7.67, 95% CI [4.718, 10.622], $p < .001$), in the model adjusted for imagery modality (model 3: Beta = 7.592, 95% CI [4.753, 10.430], $p < .001$), and in the fully adjusted model with interactions (model 4: Beta = 7.827, 95% CI [5.040, 10.615], $p < .001$). Specifically, external focus instructions predicted greater motor learning than internal focus instructions, where adopting external focus instructions resulted in a 7.827-point increase in toss accuracy during post-test. The inclusion of imagery modality variables in model 3 significantly increased the amount of variance explained. The inclusion of attentional focus in model 2 resulted in explaining 53.9% of the variance in motor learning, but when imagery modality variables were included in model 3, the variance explained increased to 60.3% (R^2 change = 0.064, $p < .001$). The inclusion of the interaction between attentional focus and imagery modality in model 4 revealed that the relationship between imagery modality and motor learning differed for each attentional focus group (change in $F = 3.00, p = .033$). The inclusion of the interaction term also significantly increased the amount of variance explained from 60.3% to 62.9% (R^2 change = 0.026, $p = .033$).

In the final model with main and interaction effects (model 4), only the internal-visual imagery modality had a significant main effect on motor learning (Beta = 0.505, 95% CI [0.077, 0.932], $p = .021$). In particular, for every one-point increase in internal-visual imagery dominance, participants scored 0.770 higher during the motor learning assessment. The main effects for external-visual imagery and kinesthetic imagery were not statistically significant (both $ps > 0.05$). Similarly, the interaction between attentional focus group and internal-visual imagery was not statistically significant ($p > .05$) (see Fig. 2a). However, these findings were superseded by the significant interactions between attentional focus group and external-visual imagery (Beta = 0.941, 95% CI [0.059, 1.824], $p = .037$) and kinesthetic imagery (Beta = -0.945, 95% CI [-1.764, -0.126], $p = .024$) for motor learning. In particular, for every one-point increase in external-visual imagery dominance, participants in the external focus group scored 0.941 points higher than the internal focus group during the motor learning assessment (see Fig. 2b). However, for every one-point increase in kinesthetic imagery dominance, participants in the external focus group scored 0.945 points lower relative to the internal focus group during the motor learning assessment (see Fig. 2c).

4. Discussion

In the present study, we examined whether differences in motor imagery modality dominance would modulate the effects of

Table 1

Means and standard deviations from the MIQ-C for the three motor imagery modality subscales of the MIQ-C.

	Imagery modality		
	Internal-visual	External-visual	Kinesthetic
All participants ($N = 138$)	21.82 $\pm$ 3.67	21.97 $\pm$ 3.87	22.55 $\pm$ 3.72
Internal focus ($n = 71$)	21.38 $\pm$ 3.43	22.38 $\pm$ 3.83	22.64 $\pm$ 3.70
External focus ($n = 67$)	22.29 $\pm$ 3.88	21.55 $\pm$ 3.88	22.46 $\pm$ 3.76

Table 2

Participants' responses to the "what were you focusing on while performing the tosses" in different attentional focus groups after acquisition.

	Internal focus group (%)	External focus group (%)
Reported external foci		
Ball	0.47	79.93
Ball trajectory	–	8.79
Bull's-eye/target	4.7	10.78
Reported internal foci		
Arm movement/trajectory	11.27	–
Arm	80.04	–
Wrist	3.05	0.50
Other foci/irrelevant foci		
Nothing/I don't know/I don't remember	0.47	–
Total	100	100

Table 3

Participants' responses to the "what were you focusing on while performing the tosses" in different attentional focus groups after the post-test.

	Internal focus group (%)	External focus group (%)
Reported external foci		
Ball	5.63	72.39
Ball trajectory	–	10.45
Bull's-eye/target	5.63	17.16
Reported internal foci		
Arm movement/trajectory	7.04	–
Arm	78.88	–
Wrist	1.41	–
Other foci/irrelevant foci		
Nothing/I don't know/I don't remember	1.41	–
Total	100	100

Table 4

Means and standard deviations from the toss accuracy in pre-test and post-test for the internal and external focus groups.

	Internal focus group (n = 71)	External focus group (n = 67)
Pre-test		
Mean	40.61	41.55
SD	10.73	11.06
Post-test		
Mean	44.29	52.68
SD	10.91	13.29

internal or external attentional focus instructions on motor learning of a ball-tossing task in 10 to 11 years old children. Overall, our results indicated a main effect of attentional focus, such that external focus produced superior motor learning compared to an internal focus. More importantly, however, the extent to which internal and external focus of attention contributed to motor learning was dependent on children's motor imagery modality dominance. Specifically, results indicated that higher levels of external-visual imagery dominance resulted in greater motor learning while adopting an external focus. In contrast, higher levels of kinesthetic imagery dominance reduced motor learning in children while adopting an external focus. These novel findings provide important contributions to the literature on the relationships between attentional focus instructions and motor imagery modalities in children, while further clarifying the role of kinesthetic, internal- and external-visual modality dominance, respectively.

The present results indicating that a higher external-visual imagery ability benefits children when adopting an external focus expands upon previous findings in adults (Bahmani, Diekfuss, & Kharestani, 2019; Sakurada et al., 2016, 2019, 2017). Specifically, prior studies suggested an external focus is most beneficial for participants with high visual imagery dominance; however, these studies did not differentiate between internal- and external-visual imagery dominance. As external-visual imagery dominance allows individuals to visualize themselves from a third-person perspective while performing specific movements (Yu et al., 2016), it is possible that an external focus capitalized on these children's unique capability to 'see' themselves attending to their desired movement outcomes in relationship to the environment. Although there was a trend such that higher internal-visual imagery dominance improved motor learning while adopting an internal focus, interestingly, the interaction between attentional focus and internal-visual imagery was not significant. It is likely that the previous research that showed a link between higher visual imagery dominance and external focus was being driven by external visual imagery ability rather than an internal visual imagery ability (Bahmani, Diekfuss, & Kharestani, 2019; Sakurada et al., 2016, 2019, 2017). However, future research is needed to test this hypothesis.

With respect to kinesthetic imagery, our results indicated that an external focus may not be particularly beneficial for children with higher levels of dominance in this modality (Sakurada et al., 2016, 2019, 2017). Although adopting an external focus still tended to be

Table 5

The results from a nested multiple regression analyses using pre-test, attentional focus group, and motor imagery modality dominance (internal-visual, external-visual, kinesthetic) to predict participants' toss accuracy at post-test (i.e., motor learning).

Variable	Model coefficients			
	Beta	95% CI	t	p
Model 1: $R^2 = 0.449$; $F = 110.626$; $p < .001$				
(Constant)	15.981	[9.683, 22.279]	5.018	<.001
Main Effect: Pre-test Score	0.789	[0.640, 0.937]	10.518	<.001
Model 2: $R^2 = 0.539$; Change in $F = 26.401$; $p < .001$				
(Constant)	12.883	[6.980, 18.786]	4.316	<.001
Main Effect: Pre-test Score	0.773	[0.637, 0.910]	11.226	<.001
Main Effect: External vs. Internal (reference) focus group	7.670	[4.718, 10.622]	5.138	<.001
Model 3: $R^2 = 0.603$; Change in $F = 7.107$; $p < .001$				
(Constant)	14.604	[9.005, 20.203]	5.160	<.001
Main Effect: Pre-test Score	0.732	[0.603, 0.861]	11.223	<.001
Main Effect: External vs. Internal (reference) focus group	7.592	[4.753, 10.430]	5.291	<.001
Main Effect: Internal-visual Imagery	0.505	[0.077, 0.932]	2.334	.021
Main Effect: External-visual Imagery	0.515	[0.104, 0.927]	2.477	.015
Main Effect: Kinesthetic Imagery	-0.430	[-0.816, -0.045]	-2.207	.029
Model 4: $R^2 = 0.629$; Change in $F = 3.000$; $p = .033$				
(Constant)	15.867	[10.141, 21.594]	5.482	<.001
Main Effect: Pre-test Score	0.706	[0.576, 0.837]	10.734	<.001
Main Effect: External vs. Internal (reference) focus group	7.827	[5.040, 10.615]	5.556	<.001
Main Effect: Internal-visual Imagery	0.770	[0.076, 1.464]	2.194	.030
Main Effect: External-visual Imagery	0.238	[-0.342, 0.818]	0.812	.418
Main Effect: Kinesthetic Imagery	-0.161	[-0.716, 0.393]	-0.576	.566
Interaction Effect: Focus Group*Internal-visual Imagery	-0.685	[-1.589, 0.219]	-1.500	.136
Interaction Effect: Focus Group*External-visual Imagery	0.941	[0.059, 1.824]	2.111	.037
Interaction Effect: Focus Group*Kinesthetic Imagery	-0.945	[-1.764, -0.126]	-2.283	.024

better than an internal focus overall, the benefit was greatly reduced and likely eliminated for children with higher values of kinesthetic imagery dominance. In other words, high kinesthetic imagery ability reduces the benefits of external focus instructions. Directing attention externally may have been difficult for children with high kinesthetic imagery dominance because external focus instructions are not well aligned with the content of kinesthetic imagery, which consists of proprioceptive awareness and aptitude towards 'feeling' their movements. That is to say, external focus instructions to "focus on the ball" is antithetical to the tendency of individuals with high kinesthetic imagery dominance. As such, this apparent incompatibility may reduce or harm motor learning while adopting an external focus of attention.

With respect to the internal focus group, higher values of kinesthetic imagery dominance also resulted in a decrease in motor learning (Fig. 2c). However, the negative effect of higher kinesthetic imagery dominance was less pronounced in the internal focus group compared to the external focus group. Overall, our findings showed that an external relative to an internal focus instruction improved motor learning for all motor imagery modalities, yet, this advantage diminished at higher levels of kinesthetic imagery ability.

Although previous research on the interaction of attentional focus and motor imagery ability reported that adopting an internal relative to an external focus of attention could be more beneficial for adults with high kinesthetic motor imagery dominance in fine motor tasks (Sakurada et al., 2016, 2019, 2017), our findings did not show any advantages for an internal relative to external focus of attention for children. It is important to note, that in the present study, the children trained on a gross motor task, whereas previous literature utilized a fine motor task. As such, the discrepancies in results may be attributable to differences in the cognitive demands between gross and fine motor control tasks (e.g., hand movement/cursor movement on screen) (Raisbeck & Diekfuss, 2015), or differences in motor imagery ability due to development of imagery modality dominance throughout childhood (Spruijt et al., 2017).

Findings from the present study may also be understood with respect to the regulation of the central nervous system (CNS). Specifically, adopting an attentional focus that is congruent with an individual's motor imagery dominance differentially regulates the CNS (Sakurada et al., 2019). For instance, when those with high visual imagery dominance adopt an external focus, and those high kinesthetic dominance adopt an internal focus, brain activity in parietal and frontal brain regions was reduced (Sakurada et al., 2019). The authors concluded that an optimal (i.e., congruent) attentional focus strategy may promote neural efficiency. However, the preliminary nature of these studies, as well as the utilization of functional near-infrared spectroscopy (which does not examine the entire cortex) warrants caution. Importantly, our participants were children within age ranges associated with heightened CNS plasticity (Gogtay et al., 2004; Peters et al., 2012; Spear, 2013; van Duijvenvoorde, Achterberg, Braams, Peters, & Crone, 2016). Thus, developmental-related CNS mechanisms may have also distinctly contributed to the present findings. However, despite future research needed to elucidate potential underlying neurophysiologic mechanisms, the present results indicate that differences in motor imagery modality dominance play a considerable role when learning a novel motor task.

The present study contained a few limitations that should be noted. First, it is unlikely that an experimental (laboratory-based) paradigm can replicate the cognitive, motor, and perceptual demands of learning motor skills in the real world. Second, the present

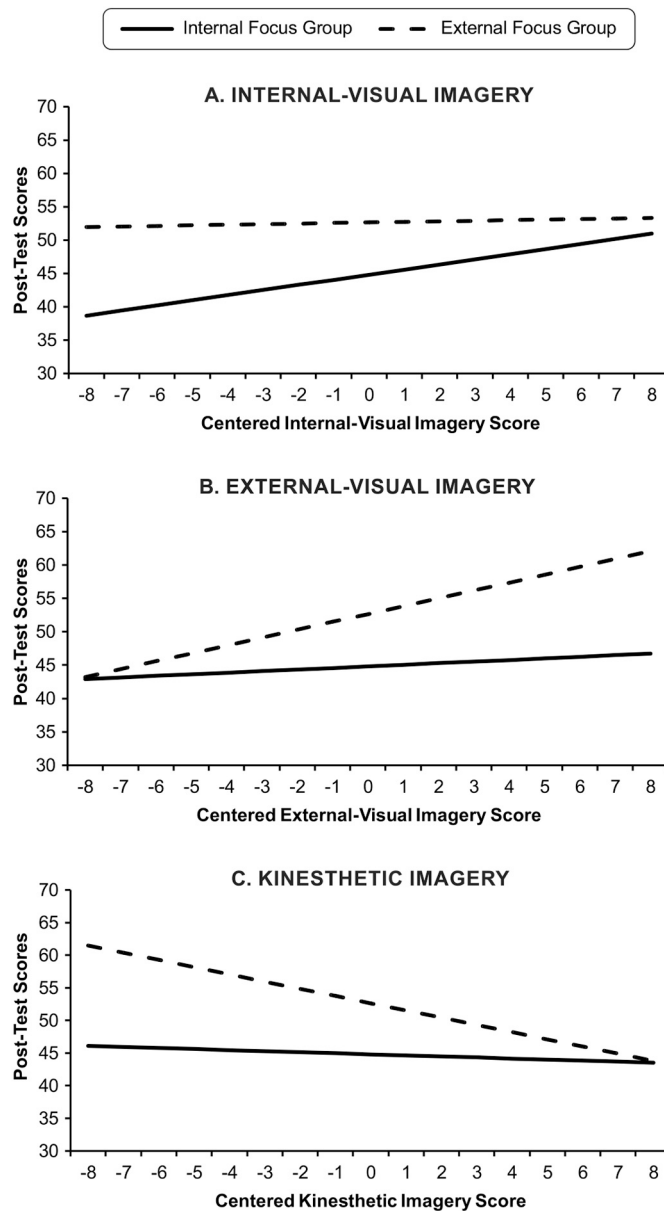


Fig. 2. Predicted values of the motor learning assessment at post-test for internal and external focus groups, at centered values of (A) internal-visual imagery, (B) external-visual imagery, and (C) kinesthetic imagery, and at the mean level of pre-test motor learning assessment values.

study utilized a relatively short acquisition (single session) and retention period (24 h) that precludes insight into whether the observed findings are retained for longer periods (e.g., weeks or months). While the present experimental design aligns with classic motor learning methodology, caution should be considered when attempting to generalize these findings, particularly due to limited ecological validity relative to real-world scenarios and the uncertainty as to whether the observed effects would be maintained for longer periods. Future research is needed in settings with higher ecological validity (Bahmani, Bahram, Diekfuss, & Arsham, 2019) and longer retention periods to determine how psychological and motor learning phenomena, such as focus of attention and motor imagery, contribute to long-term motor learning.

Notwithstanding these limitations, the present study makes valuable contributions, and may also partially explain inconclusive findings that have demonstrated that an external focus may not always be beneficial for enhancing motor performance and learning in children (Chow et al., 2014; Emanuel et al., 2008; Perreault & French, 2016; Tse & van Ginneken, 2017; van Abswoude, Nuijen, van der Kamp, & Steenbergen, 2018). Specifically, these inconsistencies in past findings may be due to the differences in childrens' motor imagery modality dominance, which was not considered in previous literature. Accordingly, we recommend that future studies incorporate motor imagery modality dominance assessments (e.g., MIQ, MIQ-C) when investigating the influence of attentional focus

on motor behavior, particularly for children. Investigating how these two variables interact, as well as their relative influence for individuals across the lifespan, could support the discovery of the optimal attentional focus strategy for personalized motor skill training.

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Declaration of Competing Interest

None.

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